FLIPKART ONLINE SHOPPING PLATFORM SALES DATA ANALYSIS

Professional Project Report | Exploratory Data Analysis

Submitted by: _____
Course / Year / Section: _____
College: _____
Academic Session: 2026–27

Dataset: 1,200 synthetic illustrative order records. This is not actual Flipkart transaction data.

Executive Summary

This project analyses 1,200 synthetic Flipkart-style order records across 2025. Delivered-order net sales total ■17,321,194; there are 1,099 delivered orders and the average delivered order value is ■15,761. Return rate is 5.8%, cancellation rate is 2.7%, average rating is 3.96/5 and average delivery duration is 4.2 days.

KPI	Result
Total orders	1200
Delivered orders	1099
Delivered net sales	■17,321,194
Average order value	■15,761
Return rate	5.8%
Cancellation rate	2.7%

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1. Introduction

E-commerce transactions create data about products, customers, discounts, payment methods and delivery outcomes. Exploratory data analysis turns these records into business insights. This project demonstrates a Flipkart-inspired analytical workflow for academic learning.

2. Problem Statement, Objectives and Scope

Problem: How can order-level data help understand sales performance, customer segments, product demand, payment preferences and fulfilment outcomes?

• Measure delivered order volume and net revenue.

• Compare sales by customer segment, city and category.

• Study payment contribution and monthly movement.

• Assess order outcomes, delivery time and ratings.

• Develop data-supported recommendations.

Scope: 1,200 illustrative orders from 1 Jan to 31 Dec 2025. Profit, inventory, seller commissions, shipping cost and repeat-customer behaviour are outside scope.

3. Dataset and Data Dictionary

The complete dataset is in the companion Excel workbook, Sales_Data sheet. One row represents one synthetic order. Currency is INR. Net_Sales_INR includes delivered orders only; returned and cancelled orders contribute zero.

Field	Description
Order_ID / Order_Date	Unique order ID and order date
City	Customer/order city
Customer_Gender / Age_Group	Illustrative customer segments
Product_Category / Product_Name	Category and example product
Quantity / Unit_Price_INR	Units and listed unit price
Discount_Rate / Order_Value_INR	Discount fraction and discounted value
Payment_Method	Payment channel
Delivery_Days / Customer_Rating	Delivery duration and rating (1–5)
Order_Status / Net_Sales_INR	Outcome and delivered-only revenue

4. Methodology and Analysis Prompt

Prompt: Act as a business data analyst. Analyse the supplied Flipkart-style transaction dataset using Python, pandas, NumPy, Matplotlib and Seaborn. Check dimensions, data types, descriptive statistics and data quality. Analyse gender, age, city, category, payment method, monthly sales, order status, delivery duration and ratings. Calculate delivered orders, net sales, average order value, return rate and cancellation rate. Produce labelled charts, interpret patterns cautiously and state limitations. Never claim synthetic data represents actual Flipkart performance.

Methods: descriptive statistics, frequency distributions, group-by aggregation, percentage calculations and monthly time-series aggregation. Tools: Python, Jupyter, pandas, NumPy, Matplotlib, Seaborn and Excel.

KPI	Definition
Delivered orders	Count of Delivered rows
Net sales	Sum of Net_Sales_INR
Average order value	Net sales ÷ delivered orders
Return rate	Returned rows ÷ all rows × 100
Cancellation rate	Cancelled rows ÷ all rows × 100

5. Customer Profile

The age band contributing the highest delivered net sales is 26-35. Gender contribution is summarised below; sales value should not be confused with customer count or individual purchasing power.

Gender	Delivered net sales (■)
Female	8,439,999
Male	8,413,943
Other	467,252

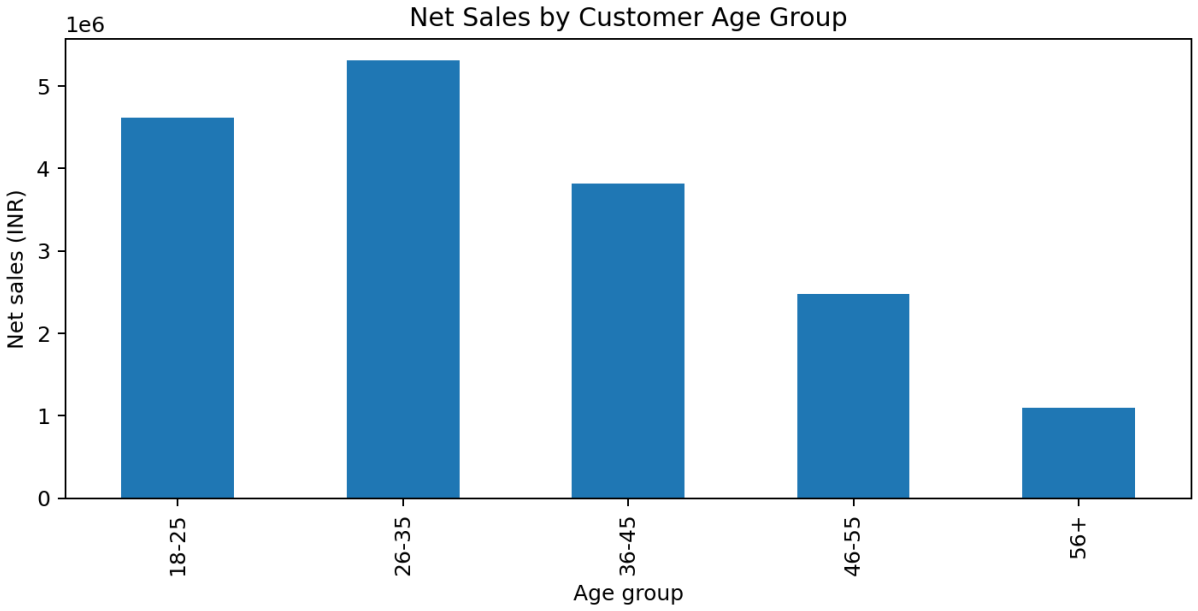


Figure 1. Net sales by age group.

6. Geographic and Product Analysis

Delhi is the highest-sales city in this generated sample. Mobiles is the leading category by delivered net sales. These results can inform regional and category-level tests, subject to validation with margin and stock data.

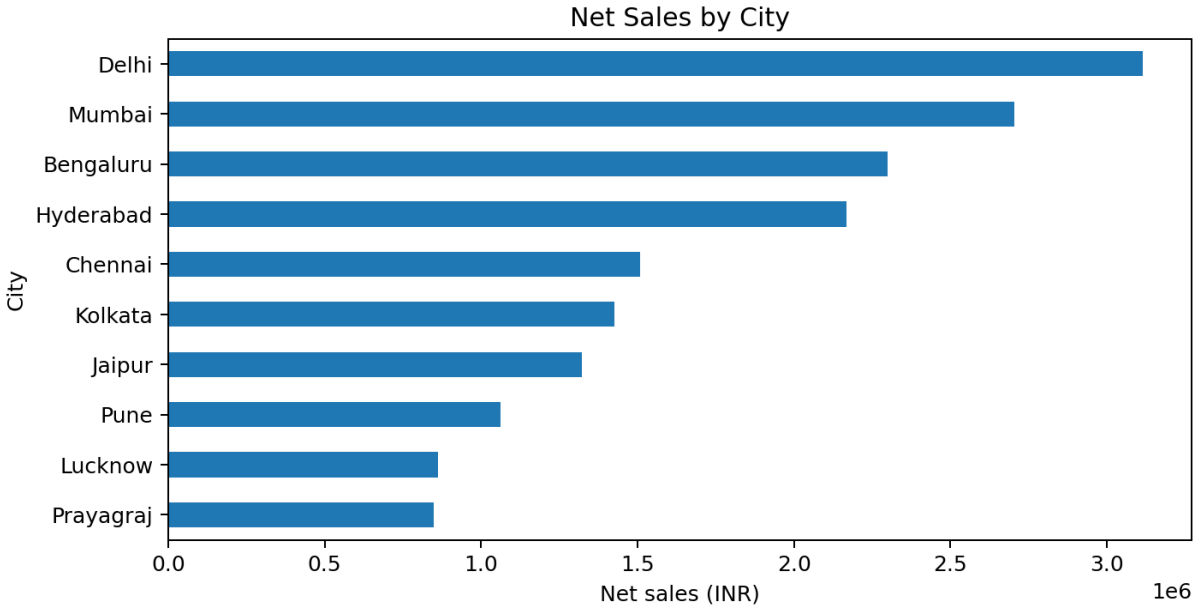


Figure 2. Net sales by city.

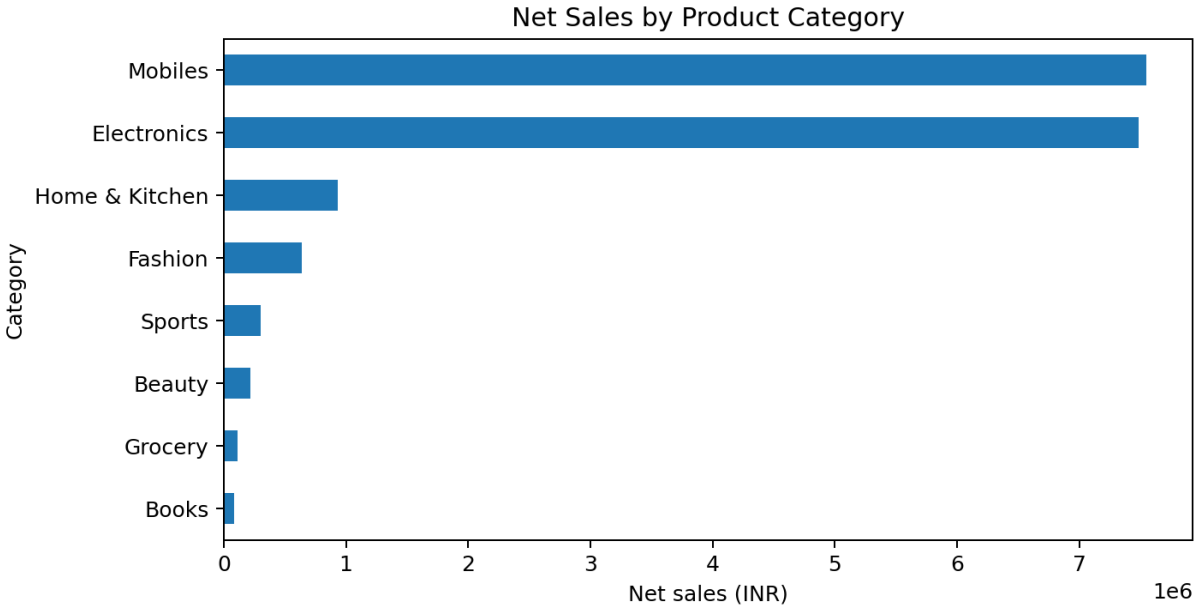


Figure 3. Net sales by product category.

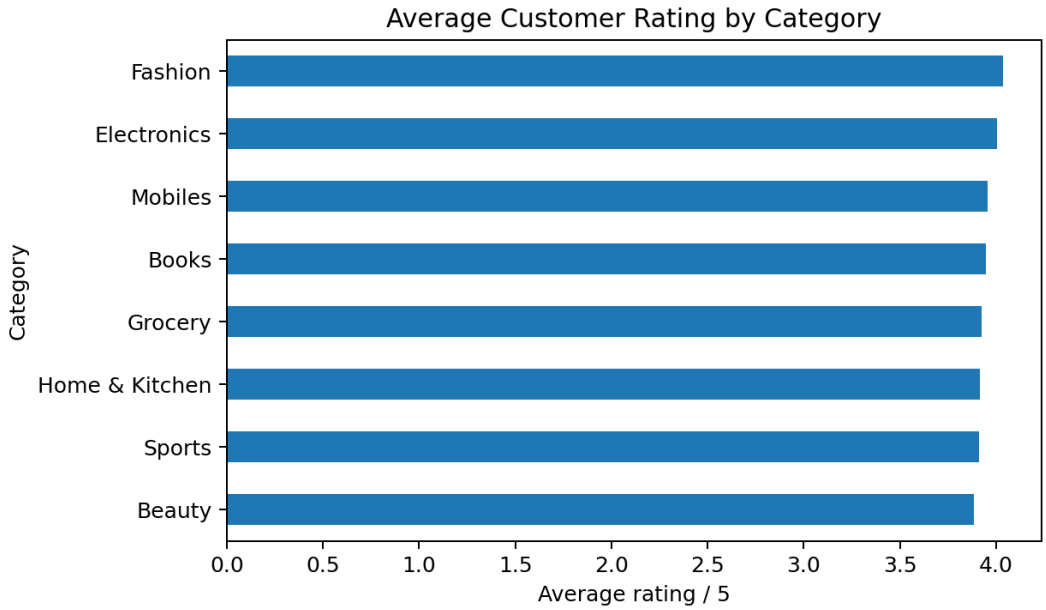


Figure 4. Average rating by category.

7. Payment, Fulfilment and Customer Experience

UPI leads delivered net sales by payment method. Returned orders represent 5.8% and cancelled orders 2.7% of all records. Average delivered-order delivery duration is 4.2 days and average rating is 3.96/5. The dataset does not explain why orders were returned or cancelled.

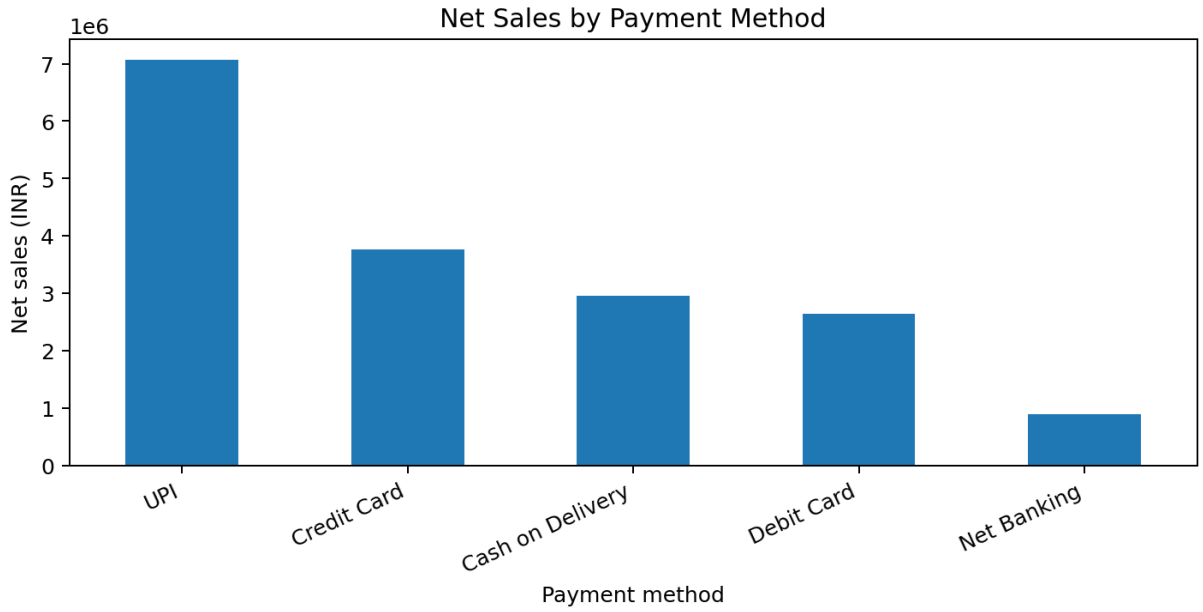


Figure 5. Net sales by payment method.

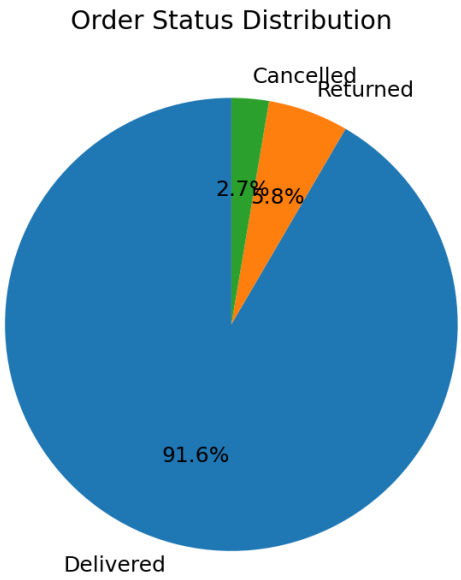


Figure 6. Order status distribution.

8. Results and Visualisations

Monthly delivered net sales peak in 2025-10 and are lowest in 2025-05 in this generated sample. The pattern is illustrative and should not be interpreted as an actual seasonal or campaign effect.

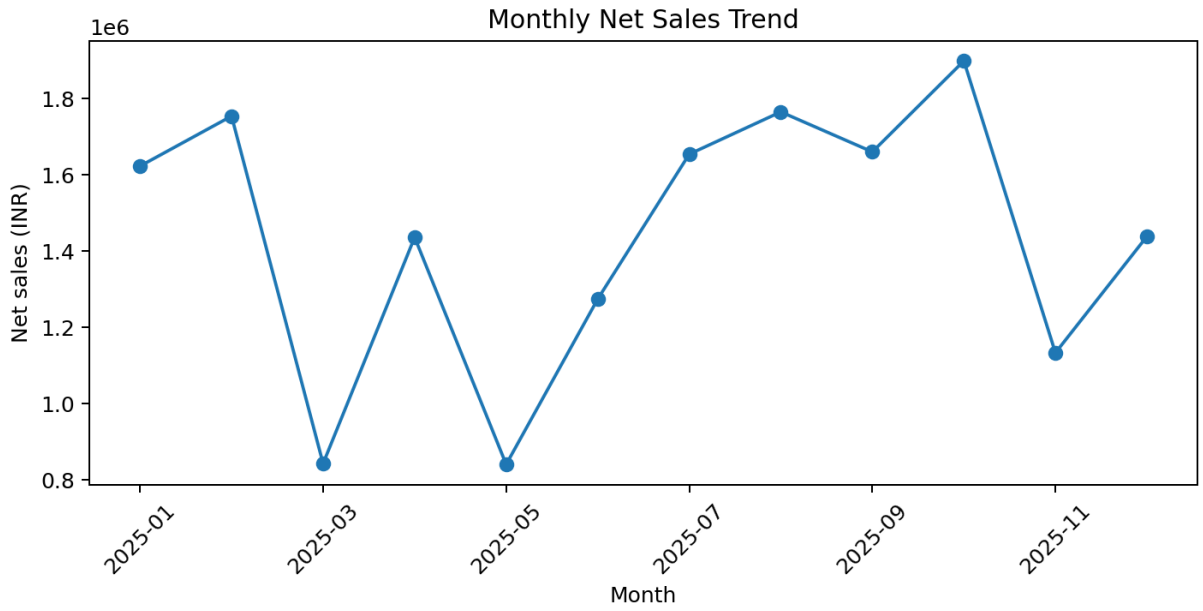


Figure 7. Monthly delivered net sales.

Result	Observed value
Top category	Mobiles
Top city	Delhi
Top payment method	UPI
Top age band	26-35
Return / cancellation	5.8% / 2.7%

9. Findings and Recommendations

• Review Mobiles for category-level merchandising while checking margin and return rates.

• Assess delivery capacity and campaign allocation in Delhi.

• Test relevant offers for the 26-35 segment and measure conversion.

• Maintain secure payment choice; monitor UPI and payment failure rates.

• Investigate return/cancellation reasons by SKU, seller and logistics partner.

• Add margin, cost, advertising and repeat-purchase data for profitability analysis.

10. Limitations and Conclusion

The dataset is synthetic, randomly generated with seed 42 and intended only for academic demonstration. It is not sourced from Flipkart and cannot support claims about actual company performance. Results are descriptive, not causal. The project demonstrates a reproducible EDA workflow and a foundation for analysis of authorised real transaction data.

Appendix A – Dataset Sheet

The companion workbook Flipkart_Sales_Dataset_Appendix.xlsx contains the complete 1,200-row Sales_Data sheet, Data_Dictionary and KPI_Summary. Preview:

Order_ID	Order_Date	City	Product_Category	Quantity	Order_Value_INR	Payment_Method	Order_Status
FK100001	2025-02-02	Chennai	Electronics	1	29620.04	Debit Card	Delivered
FK100002	2025-10-10	Chennai	Beauty	1	186.12	UPI	Delivered
FK100003	2025-08-27	Delhi	Grocery	1	1681.91	Credit Card	Delivered
FK100004	2025-06-10	Kolkata	Beauty	4	1349.0	UPI	Returned
FK100005	2025-06-08	Kolkata	Sports	1	6019.95	Credit Card	Delivered
FK100006	2025-11-10	Kolkata	Beauty	1	655.33	Debit Card	Delivered
FK100007	2025-02-01	Delhi	Books	1	584.97	UPI	Delivered
FK100008	2025-09-12	Hyderabad	Mobiles	1	8545.5	UPI	Delivered

All 1,200 records are in the Excel appendix; the report includes a compact preview to remain within the 20-page limit.

Appendix B – Reproducibility

Open Flipkart_Sales_Data_Analysis.ipynb in Jupyter or Google Colab. Upload the Excel workbook to the same working directory. Install pandas, numpy, matplotlib, seaborn and openpyxl if required. The notebook follows the section-based exploratory-analysis structure of the uploaded reference.